

Counterfactual Meta-task Augmentation for Few-shot Graph Node Classification

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Abstract

Few-shot graph node classification aims to classify unlabeled nodes with only a limited number of labeled ones. Although existing meta-learning approaches mitigate the scarcity of supervision through inter-task knowledge transfer, they remain constrained by insufficient label information and meta-task diversity under extremely weak supervision. To address this, we propose a counterfactual meta-task augmentation framework for few-shot graph learning. First, personalized PageRank is employed to extract local subgraphs around support set nodes, capturing higher-order neighborhood interactions. Then, a counterfactual variational autoencoder is designed to apply counterfactual interventions on node representations in the latent space, followed by reconstructing graph structures and node features via the decoder to generate diverse counterfactual meta-task subgraphs, upon which counterfactual meta-tasks are constructed. Finally, joint training of original and counterfactual tasks optimizes meta-parameters, enhancing the generalization and robustness of meta-learning. Extensive experiments on six public graph datasets against nine baselines demonstrate the superiority of our method under extremely weakly supervised conditions.

CCS Concepts

• **Computing methodologies** → **Semi-supervised learning settings.**

Keywords

Few-shot Learning, Counterfactual Augmentation, Meta-learning, Node Classification

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1 Introduction

Few-shot graph node classification [8] aims to predict the categories of unlabeled nodes by utilizing the graph structure and node features, especially in cases where labeled nodes are limited [42]. This task has widespread applications in areas such as drug category identification in biological molecular graphs [13] and anomaly detection in network security [2]. Traditional graph node classification methods rely on a large amount of labeled data to achieve high performance. However, in real-world scenarios, the cost of labeling is high, leading to a scarcity of labeled nodes, making it difficult for traditional methods to generalize effectively [50]. Meta-learning, as a "learning to learn" paradigm, alleviates the few-shot generalization problem through cross-task knowledge transfer [10]. In the meta-training phase, diverse few-shot tasks are constructed from relatively rich labeled meta-training classes, learning shared knowledge representations or optimization strategies across tasks. In the meta-testing phase, the model quickly adapts to new tasks with a support set. However, the effectiveness of this method relies on the diversity of meta-training tasks [12]. In situations with limited labeled nodes, the diversity of meta-tasks is restricted, and the support set struggles to capture the class distribution, leading to decreased model generalization performance. For example, ProtoNet [32] and MAML [10] perform excellently when many labeled nodes are available, but when labeled nodes are scarce, the support set fails to sufficiently represent the intra-class feature distribution, resulting in reduced model generalization [11]. Similarly, methods like TENT [40] and G-Meta [15] also suffer from limited efficiency in cross-task knowledge sharing when labeled nodes are scarce, impacting performance on target tasks [37]. Therefore, there have recently been methods focusing on data augmentation. IA-FSNC [44] addresses the problem of scarce labeled nodes through training

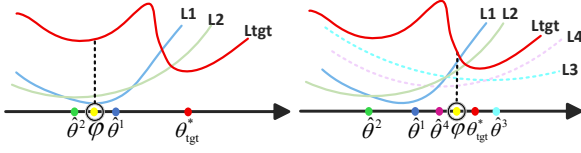


Figure 1: Illustration of the impact of meta-task diversity. For a target task L_{tgt} with optimal parameters θ^*_{tgt} , fusing the optimal parameters from a more diverse set of meta-tasks (L_1, L_2, L_3, L_4) yields an initial parameter ϕ that is closer to θ^*_{tgt} compared to fusing parameters from a less diverse set (L_1, L_2).

support set augmentation and noisy sample feature augmentation. COSMIC [43] enhances intra-class generalization ability through subgraph representation and a two-step optimized contrastive learning strategy, and utilizes a similarity-sensitive Mix-up strategy to generate hard negative classes for strengthening inter-class generalization ability. LLM4NG [48] leverages the text generation capability of LLMs to synthesize labeled nodes and edges. However, under extremely weak supervision conditions, these methods have certain limitations: the quality of pseudo-labels relied on by IA-FSNC is difficult to guarantee, COSMIC still requires a sufficient number of labeled meta-training categories to learn its contrastive features, while LLM4NG introduces powerful semantic functions, it is prone to the problem of generation randomness.

Existing works have demonstrated the effectiveness of meta-learning strategies [5, 16, 39]. But under extremely weak supervision, the construction of meta-tasks relies on a limited number of labeled nodes, resulting in insufficient task diversity and restricting the generalization ability of graph-based few-shot meta-learning [7]. This issue undermines the model’s capability to learn inductive patterns across tasks [1]. On the one hand, homogeneous tasks tend to cause overfitting, leading to poor generalization on out-of-distribution tasks [29]. On the other hand, limited task variation and imbalanced class distributions cause inconsistent gradient update directions, thereby reducing convergence efficiency and model robustness [21]. For example, methods such as ProtoNet and MAML can effectively transfer knowledge across tasks when sufficient labeled data is available. However, in label-scarce scenarios, the support set fails to capture distributional shifts, resulting in underfitting [35]. The figure shows the influence of different numbers of meta-tasks on model parameter initialization through two sub-figures. In the left figure, meta-tasks L_1 and L_2 learn to obtain parameters $\hat{\theta}^1$ and $\hat{\theta}^2$. Combining the information of these two tasks, the initial parameter ϕ is obtained, which is closer to the optimal parameter of the target task L_{tgt} than the parameter obtained by a single task. In the right figure, meta-tasks L_3 and L_4 are added, and parameter $\hat{\theta}^3$ is obtained through learning, which further enriches task diversity. This diversity makes the initial parameter ϕ closer to the optimal parameter of the target task. This indicates that insufficient task diversity constrains the generalization capacity of meta-learning, whereas increasing diversity, provided that it does not excessively introduce heterogeneity, can markedly enhance the model’s adaptability to new tasks [20].

To address the challenges of label scarcity and insufficient meta-task diversity [33, 46], we propose Counterfactual Meta-task Augmentation (CMA) for few-shot graph node classification. First, a personalized PageRank algorithm is used to extract local subgraphs of the support set nodes, capturing higher-order neighborhood interactions. Second, a counterfactual variational autoencoder (CF-VAE) is designed, applying counterfactual interventions in the latent space to reconstruct graph structures and node features, generating diverse counterfactual meta-task subgraphs. Finally, both original and counterfactual tasks are jointly optimized to improve meta-learning generalization and robustness. The contributions of this paper are as follows: (1) We introduce counterfactual augmentation into few-shot graph node classification. This method enhances the generalization ability of meta-learning under extremely weak supervision through meta-task augmentation. (2) We propose CF-VAE, which generates counterfactual meta-tasks by performing joint perturbations on both structure and features. (3) Comparative experiments are conducted on six public datasets with nine existing few-shot graph node classification methods, validating the effectiveness of the proposed method.

2 Problem Formulation

Consider a graph $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, where $\mathcal{V}$ and $\mathcal{E}$ represent the set of nodes and edges, respectively, and $\mathbf{X} \in \mathbb{R}^{|\mathcal{V}| \times d}$ is the node feature matrix with d being the feature dimension. Let C_{tr} and C_{te} represent the category sets for meta-training and meta-testing stages, respectively, such that $C_{tr} \cap C_{te} = \emptyset$, meaning that the categories in the two stages do not overlap. The meta-training stage constructs tasks using the nodes from C_{tr} , while the meta-testing stage uses nodes from C_{te} .

In the meta-learning framework, each task $T = \{S, Q\}$ consists of a support set S and a query set Q . The meta-training tasks are sampled from C_{tr} , and the meta-testing tasks are sampled from C_{te} . For an N -way K -shot setting, the support set S contains K labeled nodes randomly sampled from N classes, where each node $v \in \mathcal{V}$ and y_v is its class label. The query set Q contains the remaining nodes from the same N classes, with no overlap in nodes with S . In the meta-testing stage, the task structure is similar to the meta-training stage, but the category labels come from C_{te} .

The few-shot graph node classification problem is defined as follows: given G , the model learns to classify nodes from new categories based on a few labeled nodes. Specifically, in each task, the support set S is used to classify nodes in the query set Q , and the model learns to generalize to new categories in meta-testing tasks through meta-training tasks.

3 Method

This paper proposes a few-shot graph node classification method based on counterfactual meta-task augmentation, as shown in Figure 2, which includes three parts: meta-task subgraph construction, counterfactual meta-task generation, and counterfactual-augmented meta-learning. Meta-task subgraph construction is performed by using personalized PageRank to extract local subgraphs of the support set nodes, capturing higher-order neighborhood interactions (a). Counterfactual meta-task generation is based on the proposed CF-VAE, which applies counterfactual interventions in the latent

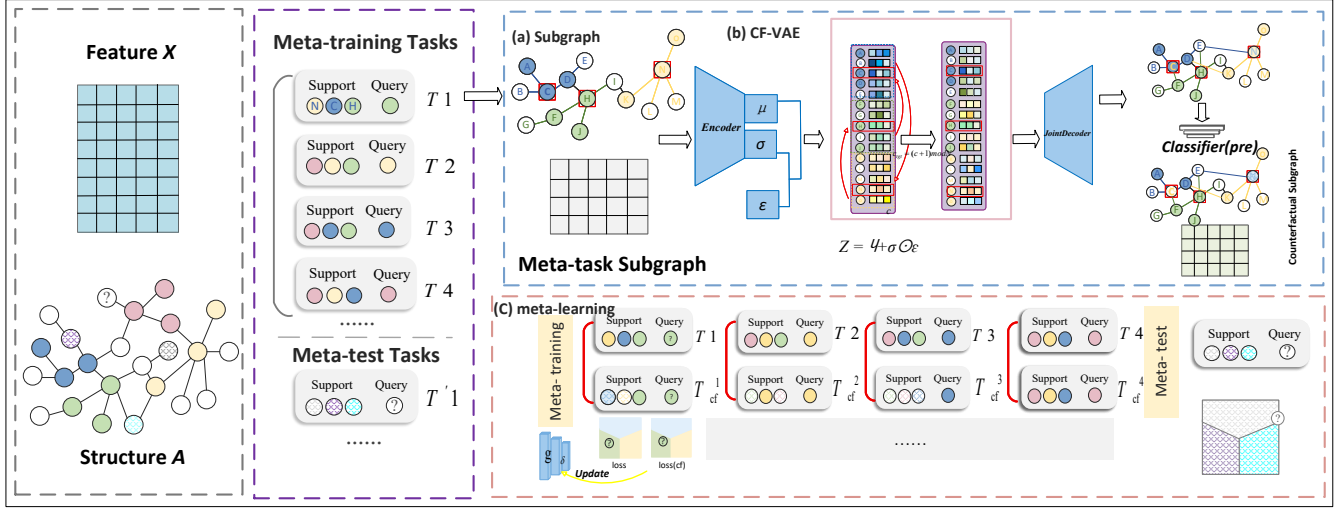


Figure 2: Few-shot graph node classification via counterfactual meta-task augmentation: (a) Extract local subgraphs from support set nodes with personalized PageRank to capture high-order neighborhood interactions; (b) The Counterfactual Variational Autoencoder (CF-VAE) intervenes in the latent space, perturbing node features and structures to generate diverse counterfactual subgraphs. Nodes in red boxes represent latent representations of target nodes, while non-red box nodes are for neighbors. Only neighbors of target nodes undergo counterfactual operations, with target nodes unchanged; (c) Jointly optimize the original and counterfactual tasks.

space to reconstruct graph structures and node features, generating diverse counterfactual meta-tasks (b). Counterfactual-augmented meta-learning improves the generalization and robustness of meta-learning by jointly optimizing the original and counterfactual tasks (c).

3.1 Meta-Task Subgraph Construction

In few-shot graph node classification, the diversity of meta-tasks is limited by the label scarcity of the original graph. Directly using the global graph for counterfactual generation would incur high computational costs. To address this, we construct local subgraphs for the support set of each meta-training task, capturing higher-order neighborhood interactions while reducing complexity. Specifically, for each meta-task $T_i = \{S_i, Q_i\}$, we generate a subgraph $G_{S_i, sub}$ for the node set V_{S_i} in the support set S_i . This subgraph is used to extract local contextual information for the nodes in the support set on the original graph G . The subgraph construction is based on Personalized PageRank, which captures higher-order dependencies between nodes through random walks. The PPR vector R_S is defined as:

$$R_S = \sum_{v_i \in S} \alpha \left(I - (1 - \alpha) \tilde{A} \right)^{-1} e_i, \quad (1)$$

where $\alpha = 0.1$ is the restart probability, $\tilde{A}$ is the normalized adjacency matrix, and e_i is the one-hot vector for node v_i . Through a Top- K selection operation, the top K nodes with the highest scores are selected from R_S and merged with the original support set nodes:

$$\mathcal{V}_{S_i, sub} = \text{Top}_K(R_S) \cup \mathcal{V}_{S_i}. \quad (2)$$

The adjacency matrix and feature matrix of the subgraph are extracted correspondingly from the original graph:

$$G_{S_i, sub} = (\mathcal{V}_{S_i, sub}, \mathbf{A}_{S_i, sub}, \mathbf{X}_{S_i, sub}). \quad (3)$$

This subgraph not only preserves local structural information but also avoids the computational burden of global calculations, providing input for subsequent counterfactual generation.

3.2 Counterfactual Meta-Task Generation

To enhance task diversity while avoiding the introduction of heterogeneity, we propose CF-VAE, which implements counterfactual interventions on features and structure in the latent space of the original graph to generate counterfactual samples that simulate whether the category of a node is likely to change if its neighborhood changes. This module compresses the subgraph representation using a variational autoencoder and applies class-guided perturbations in the latent layer, thereby reconstructing counterfactual meta-task subgraphs. By performing counterfactual interventions in the latent space [28], we can generate meta-task subgraphs in a smoother way, reducing noise and unreasonable perturbations in the original space.

First, the GCN encoder is used to extract the parameters of the latent distribution:

$$\mu, \log \sigma^2 = \text{Encoder}(\mathbf{X}_{S_i, sub}, \mathbf{A}_{S_i, sub}). \quad (4)$$

The latent representation is obtained through reparameterization sampling:

$$\mathbf{z} = \mu + \epsilon \odot \exp(0.5 \cdot \log \sigma^2), \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (5)$$

The advantage of the latent space lies in its low dimensionality and continuity, making it amenable to controllable interventions. We group the latent representations by support set categories as $\mathcal{Z}_C$, $C = 1 \dots N$, and introduce a category-specific mixing mechanism to generate counterfactual latent representations. The category feature $\mathbf{z}_j$ is weighted and fused with the latent feature $\mathbf{z}_i$ of the target category's neighborhood according to the learnable mixing weight α_c :

$$\mathbf{z}_j^{cf} = \alpha_c \cdot \mathbf{z}_j + (1 - \alpha_c) \cdot \text{Random}(\mathbf{z}_i), \quad \mathbf{z}_j \in \mathcal{Z}_C, \mathbf{z}_i \in \mathcal{Z}_{tgt}, \quad (6)$$

$$\mathcal{Z}_{tgt} = (c + 1) \bmod N, \quad (7)$$

where α_c is the learnable weight, $\alpha_c \in [0, 1]$ is a scalar parameter, and $\mathbf{z}_i$ is randomly sampled from the target category c' . This mechanism simulates cross-category feature transfer and can dynamically and adaptively learn the perturbation strength of features for different categories based on the specific task. After obtaining the counterfactual perturbation in the latent representation $\mathbf{z}^{cf}$, it is jointly decoded into the counterfactual feature representation of the nodes $\mathbf{X}^{cf}$. At the same time, the similarity between the latent variables is used to calculate the edge existence probability between node pairs, thereby generating the reconstructed adjacency matrix $\mathbf{A}^{cf}$ of the counterfactual graph:

$$\begin{aligned} \hat{\mathbf{X}}_{cf} &= \text{GCN}_3 \left(\text{ReLU} \left(\text{GCN}_2 \left(\text{ReLU} \left(\text{GCN}_1(\mathbf{Z}_{cf}, \mathbf{A}) \right), \mathbf{A} \right) \right), \mathbf{A} \right), \\ \hat{\mathbf{A}}_{ij} &= \sigma((\mathbf{z}_i^{cf})^\top \mathbf{z}_j^{cf}) \end{aligned} \quad (8)$$

CF-VAE is optimized through the following joint objective:

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + 0.5 \cdot \mathcal{L}_{\text{KL}} + 0.5 \cdot (\mathcal{L}_{\text{feat}} + \mathcal{L}_{\text{edge}}). \quad (9)$$

The reconstruction loss $\mathcal{L}_{\text{recon}}$ ensures the consistency of the reconstructed features, the KL divergence $\mathcal{L}_{\text{KL}}$ regularizes the distribution of latent variables, the feature transfer loss $\mathcal{L}_{\text{feat}}$ constrains the counterfactual features to converge towards the target category, and the edge structure loss $\mathcal{L}_{\text{edge}}$ ensures the reasonableness of the predicted edges. After generating the counterfactual subgraph, to ensure the validity of the counterfactual labels, an auxiliary classifier h_ψ (pre-trained on the subgraph) is used to predict latent label changes. This helps identify which nodes in the current counterfactual neighborhood may experience a category change, thereby enhancing the validity of counterfactual task label construction:

$$\hat{y}_i = h_\psi(\mathbf{x}_i^{cf}), \quad \forall i \in S_i. \quad (10)$$

The counterfactual label y_i^{cf} is updated only when the maximum predicted probability for node i satisfies $\max(\hat{y}_i) \geq \tau$ and its predicted category is inconsistent with the original label y_i :

$$y_i^{cf} = \arg \max \hat{y}_i \text{ if } \arg \max \hat{y}_i \neq y_i \text{ and } \max(\hat{y}_i) \geq \tau. \quad (11)$$

The selective intervention mechanism ensures the credibility of counterfactual label adjustments: only nodes that exhibit significant category drift tendencies under perturbation are updated, while the remaining nodes retain their original labels, thus avoiding the introduction of noisy samples. It is worth noting that even if the labels of nodes in the support set are not modified, counterfactual tasks can still be effectively constructed. By jointly perturbing node features and neighborhood structures, the model is placed in diverse "local environmental changes", enabling it to learn robustness to feature shifts and topological perturbations. This method does not alter

the original supervision signal, but instead enhances task diversity, thereby improving the model's adaptability to input variations.

3.3 Counterfactual-Augmented Meta-Learning

In meta-training, for a task $T = \{S, Q\}$, we generate the counterfactual support set S_{cf} and backfill the counterfactual subgraph into the original graph, achieving a balance between local perturbation and global propagation. The shared GNN encoder g_θ is used to extract embeddings:

$$\mathbf{H} = g_\theta(\mathbf{X}, \mathbf{A}), \quad \mathbf{H}_{cf} = g_\theta(\mathbf{X}_{cf}, \mathbf{A}). \quad (12)$$

Compute class prototypes:

$$\mathbf{c}_k = \frac{1}{|S_k|} \sum_{i \in S_k} \mathbf{h}_i, \quad \mathbf{c}_k^{cf} = \frac{1}{|S_k|} \sum_{i \in S_k} \mathbf{h}_i^{cf}. \quad (13)$$

Algorithm 1 Learning Process of CMA

Input: Graph $G = (\mathcal{V}, \mathcal{E}, \mathbf{X})$, meta-training tasks $T_{\text{train}} = \{S, Q\}$, meta-test tasks $T_{\text{test}} = \{S, Q\}$, number of episodes T .

Output: Predicted labels for Q nodes in meta-test tasks.

// Meta-training phase

- 1: Initialize encoder GNN_θ , counterfactual generator CF-VAE_ϕ ;
 - 2: **for** $t = 1$ to T **do**
 - 3: Sample meta-training task $T_{\text{train}} = \{S, Q\}$ from C_{tr} ;
 - 4: Extract subgraphs for support nodes in T_{train} ;
 - 5: Generate counterfactual support set T_{train}^{cf} via CF-VAE_ϕ ;
 - 6: Compute loss on original task loss_{ori} and counterfactual task loss_{cf} ;
 - 7: Update GNN_θ via joint loss $\text{loss} = \text{loss}_{\text{ori}} + \alpha \cdot \text{loss}_{cf}$;
 - 8: **end for**
 - // Meta-test phase**
 - 9: Fine-tune a simple classifier based on S using L_{CE} .
 - 10: Evaluate classifier on Q .
-

Compute the probability distribution for the query samples in both meta-tasks and counterfactual meta-tasks:

$$\begin{aligned} p(y_j = k \mid \mathbf{h}_j) &= \frac{\exp(-d(\mathbf{h}_j, \mathbf{c}_k))}{\sum_{k'} \exp(-d(\mathbf{h}_j, \mathbf{c}_{k'}))}, \\ p^{cf}(y_j = k \mid \mathbf{h}_j) &= \frac{\exp(-d(\mathbf{h}_j, \mathbf{c}_k^{cf}))}{\sum_{k'} \exp(-d(\mathbf{h}_j, \mathbf{c}_{k'}^{cf}))}, \end{aligned} \quad (14)$$

where $d(\cdot, \cdot)$ is the Euclidean distance, the original contrastive loss and the counterfactual contrastive loss are:

$$\begin{aligned} \mathcal{L}_{\text{orig}} &= - \sum_{(h_j, y_j) \in Q} \log p(y_j \mid h_j), \\ \mathcal{L}_{cf} &= - \sum_{(h_j, y_j) \in Q} \log p^{cf}(y_j \mid h_j). \end{aligned} \quad (15)$$

Joint optimization:

$$\mathcal{L}_{\text{combined}} = \mathcal{L}_{\text{orig}} + \alpha \mathcal{L}_{cf}. \quad (16)$$

Parameter update is:

$$\theta \leftarrow \theta - \eta \nabla_\theta \mathcal{L}_{\text{combined}}. \quad (17)$$

Table 1: Overall few-shot node classification results of CMA and baselines under different settings (accuracy in %).

Dataset	DBLP						Amazon-E					
Setting	5-way 3-shot			10-way 3-shot			5-way 3-shot			10-way 3-shot		
# Labels per Class	5	10	20	5	10	20	5	10	20	5	10	20
PN	49.4±3.2	51.9±3.1	53.3±3.9	36.3±3.8	38.5±2.8	40.2±3.9	51.6±2.3	52.2±2.3	53.8±2.3	36.7±3.0	38.2±2.0	41.3±3.9
MAML	50.9±3.1	51.8±1.8	56.1±2.1	39.4±2.3	44.3±2.0	45.4±3.1	48.8±2.4	49.4±3.3	53.9±2.7	39.0±3.2	40.3±3.2	41.5±3.2
G-Meta	59.8±3.3	61.8±3.5	63.3±4.1	44.9±2.9	51.0±3.4	52.9±3.6	53.4±2.2	55.7±3.6	56.6±3.2	39.6±4.1	41.9±3.0	45.6±4.3
GPN	58.6±3.8	62.5±2.8	66.9±4.3	50.6±3.9	52.7±2.4	54.6±3.4	56.0±4.1	60.7±4.7	63.0±2.3	42.1±4.8	45.8±3.3	52.1±4.8
RALE	64.7±4.1	66.9±4.7	67.9±4.0	51.3±4.2	55.0±3.2	56.9±4.0	60.4±4.5	64.0±4.8	66.1±4.5	47.8±4.4	48.6±4.8	52.4±3.3
X-FNC	70.1±4.0	75.5±3.5	76.8±3.3	57.2±3.4	63.6±3.3	65.8±3.1	69.9±3.9	72.8±3.4	76.0±4.8	49.2±4.1	51.5±2.8	56.3±3.4
TENT	75.5±3.5	77.4±3.1	77.8±3.5	65.7±4.1	65.6±3.5	66.5±2.1	70.3±4.3	71.6±3.2	72.5±3.8	61.4±3.9	62.9±3.3	62.4±3.4
COSMIC	74.5±3.2	78.5±2.5	80.6±6.6	62.9±3.2	64.7±2.8	68.7±4.1	69.3±3.2	73.9±3.2	75.6±2.8	60.5±2.6	64.7±3.6	66.8±3.5
NAQ	75.5±2.1	77.4±3.8	81.2±2.9	64.7±3.2	65.2±2.5	67.6±3.7	68.3±2.3	75.6±3.5	75.9±2.7	62.5±3.9	65.7±3.4	65.2±2.6
CMA (our)	77.3±2.1	81.4±1.2	82.1±2.8	67.1±2.5	68.7±3.2	69.4±2.9	72.2±4.1	77.4±2.5	76.3±1.8	64.2±2.2	66.3±3.4	69.9±2.6

Dataset	Cora-full						Amazon-C					
Setting	5-way 3-shot			10-way 3-shot			5-way 3-shot			10-way 3-shot		
# Labels per Class	5	10	20	5	10	20	5	10	20	5	10	20
PN	45.5±2.7	48.1±3.6	48.9±3.8	28.2±3.8	31.6±3.3	34.4±2.5	67.5±3.3	70.6±2.9	72.2±3.3	60.5±2.6	63.2±3.8	65.0±2.5
MAML	46.9±2.6	48.6±3.0	49.2±2.7	32.7±2.5	33.2±2.3	35.8±1.9	60.1±3.9	65.1±3.2	70.3±2.9	58.7±3.8	60.5±3.3	63.4±2.9
G-Meta	57.7±3.9	58.7±3.6	59.8±2.6	41.7±3.3	42.0±3.0	43.8±2.7	70.7±3.2	75.8±3.1	78.6±2.6	62.3±2.3	65.3±2.6	69.0±3.0
GPN	54.6±2.8	55.2±3.6	57.7±4.2	38.4±2.8	40.2±2.9	42.0±4.5	68.3±2.5	69.5±2.7	73.6±3.1	60.0±3.5	63.8±2.1	64.3±2.1
RALE	58.2±2.8	59.3±4.1	63.1±3.9	38.1±4.2	43.4±2.8	44.0±4.5	75.6±3.7	77.5±4.3	75.3±2.5	59.3±2.9	61.3±4.7	68.3±3.9
X-FNC	62.9±4.5	68.0±3.7	69.2±4.6	43.7±4.8	45.6±4.4	47.7±4.5	73.6±3.5	74.5±3.9	71.8±4.3	60.8±4.1	61.5±3.5	59.5±3.1
TENT	61.2±2.7	65.5±2.0	64.5±3.6	52.5±4.6	52.5±3.5	52.5±3.7	79.9±3.9	81.7±3.7	81.0±4.2	71.2±3.2	72.3±3.8	69.9±3.1
COSMIC	63.5±3.7	65.6±3.2	70.6±2.3	55.4±2.1	53.9±2.7	56.7±3.6	78.6±3.8	82.3±3.6	85.6±2.6	69.5±2.2	74.9±3.9	76.7±3.6
NAQ	63.5±2.9	66.1±3.8	71.2±3.2	52.1±2.4	54.7±3.6	56.9±3.8	83.6±3.2	85.1±2.7	86.9±3.5	73.3±3.9	76.8±3.1	78.6±2.1
CMA (our)	67.4±2.2	68.8±3.8	73.2±2.7	58.3±2.5	55.7±3.3	59.8±2.6	82.6±2.2	85.3±1.9	87.4±3.4	73.3±3.5	78.1±1.7	79.2±2.6

Dataset	CiteSeer						ogbn-arxiv					
Setting	2-way 3-shot			2-way 1-shot			5-way 3-shot			10-way 3-shot		
# Labels per Class	3	5	10	3	5	10	50	100	200	50	100	200
PN	51.3±3.3	52.7±3.5	54.2±3.3	46.8±3.9	47.3±4.5	51.4±4.2	39.1±2.5	40.8±3.7	42.6±3.1	23.1±3.6	24.4±3.1	27.7±3.5
MAML	47.4±3.7	50.9±4.1	53.2±3.2	46.5±3.9	50.6±3.5	51.7±3.2	41.0±2.4	41.9±1.9	43.1±3.4	23.2±2.1	25.4±3.1	28.0±3.2
G-Meta	57.6±3.7	60.7±3.1	61.4±3.7	50.3±3.8	51.5±3.6	54.3±4.2	43.5±3.6	44.7±2.9	46.5±4.4	27.4±4.4	29.0±2.8	29.9±2.5
GPN	52.7±3.9	56.4±2.9	58.2±3.8	53.5±2.9	53.9±2.4	55.7±3.1	46.6±3.4	47.1±3.9	48.4±2.9	26.1±2.6	30.9±3.6	33.5±3.5
RALE	57.4±3.0	60.5±3.1	62.5±4.6	54.2±3.1	55.7±3.5	56.6±3.9	49.3±3.0	51.4±3.9	52.5±4.6	30.4±2.5	31.7±3.3	33.9±4.8
X-FNC	57.4±4.1	58.6±3.1	60.2±4.1	52.6±3.6	54.3±4.1	55.6±3.4	54.6±2.6	56.7±4.0	58.7±4.1	33.3±3.9	35.7±4.4	39.8±2.4
TENT	63.5±2.9	62.2±3.5	69.3±3.6	56.9±3.2	53.9±4.1	54.7±3.1	51.6±3.4	51.7±3.6	55.4±3.1	32.5±2.5	34.1±4.2	36.8±3.0
COSMIC	61.5±3.0	65.4±3.3	68.6±3.2	60.1±2.1	59.3±3.9	62.3±2.6	53.1±3.4	55.9±2.9	54.2±2.5	34.6±3.8	34.8±2.3	38.3±2.1
NAQ	63.8±2.4	66.4±3.3	67.2±2.2	65.7±3.8	64.9±2.6	66.9±2.5	51.4±2.7	56.7±3.8	58.6±3.2	32.6±2.8	35.9±3.4	39.6±2.2
CMA (our)	64.2±2.8	67.4±4.6	70.2±2.3	64.4±3.5	65.5±4.2	67.7±3.9	53.7±2.3	56.9±3.8	57.6±2.4	33.6±4.2	36.2±2.5	37.2±2.7

At the same time, CF-VAE updates the strategy during training, enhancing the model’s sensitivity to perturbations through counterfactual tasks, thereby improving generalization. In the meta-testing phase, only the original losses are used for evaluation to ensure fairness.

4 Experimental Analysis

Datasets: To validate the effectiveness of the proposed method, we evaluate it on six benchmark graph datasets, covering various types of graphs such as citation networks, collaboration networks,

Table 2: Dataset Statistics and Settings.

Dataset	#Nodes	#Edges	#Features	Class Split
Amazon-E	42,318	43,556	8,669	90/37/40
DBLP	40,672	288,270	7,202	80/27/30
Cora-full	19,793	65,311	8,710	25/20/25
Amazon-C	24,919	91,680	9,034	40/17/20
CiteSeer	3,327	9,228	3,703	2/2/2
ogbn-arxiv	169,343	1,166,243	128	15/5/20

and product co-occurrence networks. Specifically, the datasets include Cora-full [4], an extended academic citation network where nodes represent papers, edges represent citation relationships, and categories represent research fields; CiteSeer [45], a scientific literature citation network where node categories represent paper topics; DBLP [34], an academic collaboration network where nodes represent authors, edges represent co-authorship relationships, and categories represent research fields; ogbn-arxiv [14], a large-scale paper citation network where node features are embeddings of titles and abstracts, and categories represent arXiv topics; Amazon-Electronics and Amazon-Clothing [27], which are co-purchase networks based on Amazon products where nodes represent products, edges represent co-purchases, and categories represent product types. The dataset statistics are shown in Table 2, where the category split is provided for meta-training/validation/test categories.

Experimental Setup and Evaluation: To comprehensively evaluate the generalization ability of the proposed method, we design an extremely weak supervision setting and test across multiple datasets. In DBLP, Amazon-Electronics, Cora-full, and Amazon-Clothing, the number of labeled nodes per meta-training class is limited to 5, 10, and 20. In the smaller CiteSeer dataset, the per-class labeled counts are 3, 5, and 10; in the larger ogbn-arxiv dataset, they are 50, 100, and 200. For label proportions, on ogbn-arxiv the labeled nodes constitute 0.4%, 0.8%, and 1.7% of all nodes, on DBLP, Amazon-Electronics, Cora-full, and Amazon-Clothing they are approximately 1%, 2%, and 4%, on CiteSeer they are 0.1%, 0.3%, and 0.6%. During experiments, meta-training tasks are formed by randomly sampling a limited number of labeled nodes from the meta-training classes to build support and query sets; meta-test tasks are sampled from the test classes. The primary evaluation metric is the average classification accuracy over meta-test tasks. The auxiliary MLP classifier is pre-trained on subgraphs for 100 epochs with a confidence threshold of 0.7 ($\tau = 0.7$). For each node, we extract the Top-80 nodes to form a subgraph, and the mixing weight is set to $\alpha = 0.5$. The GNN hidden dimension is 64, and the dropout rate is 0.6. We run 5000 meta-training episodes, use 50 meta-validation tasks and 500 meta-test tasks, and use 10 query samples per task. The experiments were conducted in an environment running Python 3.7.10 and PyTorch 1.8.1 on a Windows 10 Professional system equipped with an Intel i7-10700 CPU, 16 GB of RAM, and an NVIDIA GeForce RTX 3090 GPU.

4.1 Baselines

We compare the proposed method with nine representative few-shot graph node classification methods, including ProtoNet, which

achieves cross-task classification through prototype embeddings, MAML, which learns a fast adaptation initialization, G-Meta, which uses local subgraphs for meta-gradient optimization, GPN, which integrates node importance with prototype comparison, RALE [26], which captures relative and absolute position embeddings to improve transfer; X-FNC [41], a data-augmentation-based meta-learning method that generates pseudo-labels through an information bottleneck to expand supervisory signals, TENT, which reduces task variance through multi-level adaptation, COSMIC, another data-augmentation-based meta-learning method that generates additional hard-to-classify categories through similarity-based subgraph mixing to enhance inter-class distinguishability, and NAQ, which proposes an unsupervised episode generation approach.

4.2 Experimental Results

In the extremely weakly supervised few-shot node classification task, we compare the proposed method with nine mainstream baseline methods (Table 1). The experimental results show that the method outperforms most baseline methods across all datasets and settings, verifying its effectiveness and robustness under extremely weak supervision. As task complexity increases (e.g., from 5-way 3-shot to 10-way 3-shot), most baseline methods suffer from significant performance degradation, while our method benefits from counterfactual meta-tasks generated by the CF-VAE and the selective label updating based on the confidence threshold of the auxiliary classifier, effectively suppressing noise caused by unreliable perturbations and maintaining a slower performance degradation trend in the extended meta-tasks.

Furthermore, when the number of samples per class (K) in the support set increases from 3 to 10, most methods show limited improvement, indicating their heavy reliance on labeled data. In contrast, our method consistently maintains a leading performance with more stable improvement. This is mainly due to: 1) the counterfactual subgraphs being integrated into the original graph provide richer supervision signals during meta-training, and 2) the enhanced task diversity through counterfactual meta-task construction alleviates the performance bottleneck caused by label scarcity.

4.3 Ablation Study

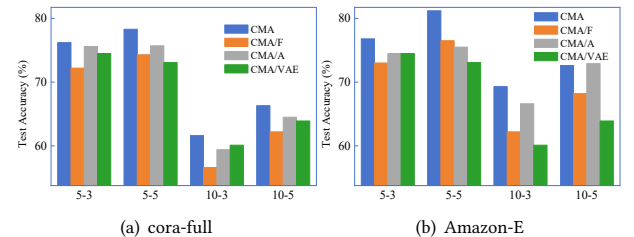


Figure 3: The ablation study of CMA on the Cora-full and Amazon-E datasets under different N -way K -shot settings.

We validate each module on Cora-full and Amazon-E by comparing the full model (CMA) with three variants—CMA/F (removes the counterfactual feature module, keeping features identical to the

original subgraphs), CMA/A (removes the counterfactual structure module, keeping adjacencies identical), and CMA/VAE (replaces the CF-VAE with a vanilla VAE). As shown in Fig. 3, CMA consistently outperforms all variants, confirming the necessity of the CF-VAE in the overall framework. The drop of CMA/F is most pronounced, especially as the task scale increases (e.g., $N = 10$), indicating that feature perturbations generated by the counterfactual mechanism are crucial for constructing discriminative counterfactual tasks and sharpening boundaries between semantically similar classes. CMA/A performs reasonably on small tasks but degrades notably on the more complex Amazon-E, demonstrating that structural perturbations introduced by the CF-VAE bring useful diversity and enhance topological discrimination. Replacing the CF-VAE with a vanilla VAE (CMA/VAE) also leads to a substantial performance drop. This is mainly due to the loss of controlled and semantically meaningful perturbations: unlike the CF-VAE, a vanilla VAE captures general variations without considering the counterfactual distribution, resulting in less informative task augmentations that may even mislead training. The perturbations generated by the CF-VAE produce feature and structural changes aligned with plausible alternative scenarios. Overall, CMA effectively mitigates the limited meta-task diversity and enhances model robustness through the CF-VAE and its coordinated feature and structure perturbations.

4.4 Parameter Analysis

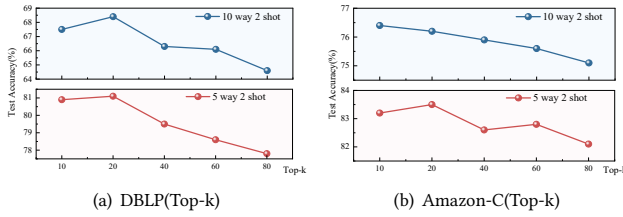


Figure 4: Accuracy under different top-K selections on DBLP (left) and Amazon-C (right) datasets.

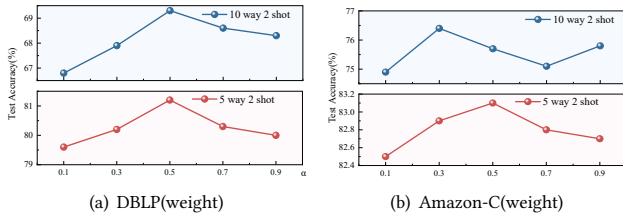


Figure 5: Accuracy under different counterfactual task loss weights on DBLP (left) and Amazon-C (right) datasets.

Based on the experimental results on the DBLP and Amazon-C datasets, as shown in Figure 4 and 5, we can observe the performance trends of the model under different subgraph scales (top-K) and counterfactual loss weights. Firstly, as shown in the left subfigure, when the subgraph scale is small (e.g., top-20), the test accuracy

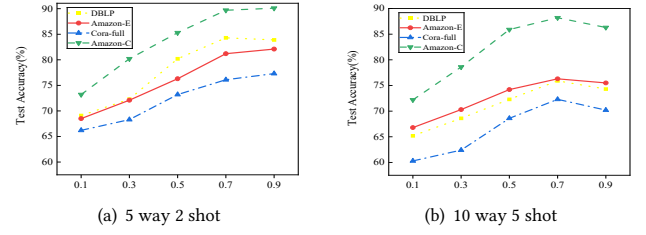


Figure 6: The figure shows the overall accuracy of the model on four datasets in the few-shot graph node classification tasks of "5-way 2-shot" (a) and "10-way 5-shot" (b), respectively, under different MLP thresholds (0.1, 0.3, 0.5, 0.7, 0.9).

is generally higher under both the 5-way and 10-way settings. This indicates that smaller subgraphs can more accurately capture the neighborhood structure information that is crucial to the target node, thereby avoiding the introduction of redundant nodes that may contain irrelevant information, and improving the discriminative capability of the task. Secondly, as shown in the right subfigure, as the counterfactual loss weight increases, the model's accuracy initially improves and peaks around weight = 0.5, then starts to decline. This suggests that a properly tuned counterfactual loss weight can effectively enhance task diversity while preserving the semantic information of the original task, thereby improving the model's robustness and generalization ability. However, when the weight becomes too large, the counterfactual task may cause semantic drift, leading to performance degradation.

Subsequently, we conducted hyperparameter experiments on the MLP confidence threshold τ , as shown in Fig. 6. The results indicate that the threshold setting has a significant impact on counterfactual pseudo-label updates and model training: when the threshold is too low (τ small), the auxiliary classifier includes low-confidence nodes in the label updates, introducing a large number of noisy labels and disrupting training, when the threshold is too high (τ large), only a few nodes meet the update condition, which reduces noise but limits the diversity of counterfactual tasks and restricts improvements in model generalization. To balance this trade-off, we propose a selective label update mechanism that updates only nodes meeting the confidence threshold. Experimental results show that $\tau = 0.7$ serves as a stable and effective operating point on the current benchmark, suppressing noise while maintaining the diversity of counterfactual tasks, thereby facilitating model training. Although this simple threshold strategy is effective, its static nature has limitations. Future work could explore more sophisticated strategies to dynamically balance noise control and task diversity, further enhancing model robustness and adaptability.

4.5 Time Complexity Analysis

The time complexity analysis of the algorithm in this paper is as follows: In the subgraph extraction stage, its time complexity is $T_{\text{subgraph}} = O(|S| \cdot k)$, where $|S|$ represents the size of the support set and k is the neighborhood truncation threshold. The time complexity of the counterfactual generator:

$$T_{\text{CF-VAE}} = O\left(|\mathcal{V}| \left(\sum_{j=1}^{L_{\text{vae}}-1} h_j' h_{j+1}' + 2 \cdot h_{\text{vae}} z \right) + |\mathcal{E}| \sum_{j=1}^{L_{\text{vae}}} h_j' \right), \quad (18)$$

where the first part corresponds to the encoder complexity of the VAE, and the second part corresponds to the decoder complexity. L_{vae} represents the number of VAE network layers, h'_j is the dimension of each layer, and z is the dimension of the latent variable. The time complexity of the graph neural network encoder:

$$T_{\text{GNN}} = O(|\mathcal{E}| \sum_{i=1}^L h_i + |\mathcal{V}| \left(\sum_{i=1}^L h_i h_{i+1} + h_L C \right)), \quad (19)$$

where L is the number of GNN layers, h_i is the dimension of each layer, and C is the number of output categories. And the total time complexity is $T_{\text{total}} = T_{\text{subgraph}} + T_{\text{CF-VAE}} + T_{\text{GNN}}$. By separating the costly global training from the low-cost local augmentation, we successfully control the resulting computational overhead, enabling the entire method to be effectively applied to large-scale graphs, while the performance bottleneck remains the GNN itself.

4.6 Visualization Analysis

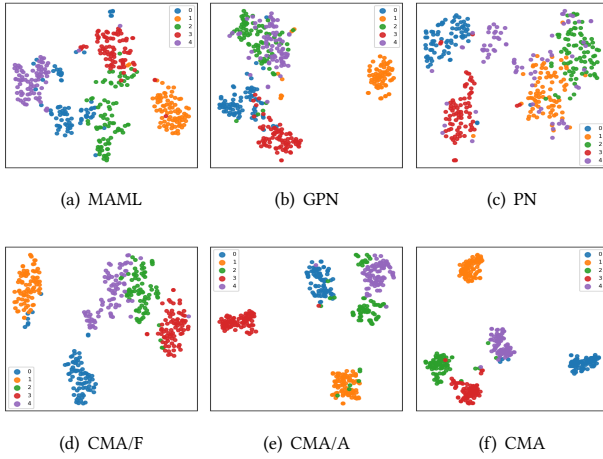


Figure 7: Visualization results on the DBLP dataset under the 5-way 2-shot setting during the test phase. Subfigures (a)–(c) represent different baseline models, while (d)–(f) show model variants and our proposed method.

Figure 7 illustrates the t-SNE visualization of query set embeddings under the 5-way 2-shot setting on the DBLP dataset. Subfigures (a)–(c) correspond to baseline models (MAML, GPN, PN), while (d)–(f) show results of our model variants (CMA/F, CMA/A) and the proposed CMA. We observe that baseline models often produce overlapping embedding clusters—for instance, some classes in MAML’s visualization (a) are intertwined—and their intra-class distributions are scattered. This directly reflects the weak discriminability of features learned by baselines, as they struggle to distinguish between different classes or maintain consistency within the same class. In contrast, our CMA model (f) exhibits significantly enhanced inter-class separability and intra-class consistency: different classes form clear, non-overlapping clusters, and instances of the same class aggregate tightly around cluster centers. Even model variants (d, e) show slight improvements over baselines but fall short of CMA. This improvement stems from the counterfactual meta-tasks introduced by CMA. These tasks generate diverse,

Table 3: Average Rankings (Friedman Test) on DBLP. CMA consistently ranks first. Sets A/B denote 20/10 labeled nodes per class.

Method	5-way 3-shot		10-way 3-shot	
	Set A	Set B	Set A	Set B
ProtoNet	9.50	9.50	9.33	9.50
MAML	9.50	9.33	9.33	9.00
G-Meta	7.00	7.00	7.00	7.33
GPN	7.67	7.33	7.17	7.00
RALE	5.92	5.83	6.67	5.67
X-FNC	4.25	4.67	5.17	5.33
TENT	3.25	4.00	3.33	4.67
COSMIC	3.58	3.50	3.00	2.67
NAQ	3.00	2.50	2.58	2.33
CMA (Ours)	1.33	1.33	1.42	1.50

informative training signals by perturbing node features and structures, prompting the model to learn more robust representations. Unlike baselines that rely only on limited original task data, CMA’s augmented signals enable better generalization on unseen few-shot tasks in the meta-test phase.

4.7 Statistical Significance Analysis

Based on the Friedman test results in Table 3, CMA achieves the lowest average ranking across all evaluation settings on the DBLP dataset, indicating superior overall performance with strong statistical robustness under multiple tasks and configurations. Under both 5-way 3-shot and 10-way 3-shot scenarios, CMA ranks first in Set A and Set B, maintaining a clear margin over the second-best methods even in low-label settings, which demonstrates its robustness. In contrast, traditional meta-learning methods (e.g., ProtoNet and MAML) perform poorly, while existing graph meta-learning approaches show moderate improvements. Overall, these results confirm that the performance gains of CMA are consistent across tasks rather than driven by a small number of favorable cases.

5 Conclusion

This paper tackles few-shot graph node classification under extreme weak supervision by introducing a counterfactual meta-task augmentation framework. CF-VAE generates diverse counterfactual subgraphs to enrich training, and the model jointly optimizes losses on original and counterfactual tasks. Experiments on six benchmarks show consistent gains over strong baselines, validating the approach. Future work will explore adaptive counterfactual generation and dynamic task construction to further improve generalization.

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A Related Work

A.1 Semi-supervised graph node classification

Semi-supervised graph node classification leverages the structural and feature information of a few labeled nodes together with many unlabeled nodes to predict the latter’s categories. Existing methods addressing label scarcity can be divided into three classes: graph regularization and propagation, graph neural networks (GNNs), and data augmentation. Graph regularization and propagation methods assume neighboring nodes share labels and exploit label–structure correlations via explicit constraints or diffusion. The label propagation (LP) algorithm [51] spreads known labels to unlabeled nodes through iterative diffusion, while the graph regularization (GR) framework [3] adds a Laplacian smoothing term to enforce feature–label consistency. These approaches are interpretable and backpropagation-free but capture only shallow neighborhood information. GNN-based methods jointly encode structure and features through end-to-end aggregation. Graph Convolutional Network (GCN) [18] smooths neighbor features via a first-order spectral approximation, and APPNP [19] integrates personalized PageRank with feature mixing to reduce information dilution. However, they generalize poorly under extremely weak supervision. Data augmentation methods exploit unlabeled nodes by transforming graph data. GraphMix [36] interpolates node features and enforces prediction consistency between original and augmented nodes. Despite their progress, these methods still struggle in few-shot scenarios, motivating meta-learning approaches to transfer knowledge across tasks under extreme label sparsity.

A.2 Graph Few-shot Learning Based On Meta-learning

Meta-learning [10] enables rapid adaptation of the model to new tasks with few samples by learning commonalities across tasks [31]. Methods addressing the few-shot problem can be classified into three categories: model initialization-based methods [22], metric learning-based methods [38], and optimization strategy-based methods [30]. Model initialization-based methods learn transferable parameter initialization that supports quick adaptation. MAML, proposed by Finn et al., learns a universal initialization through meta-optimization, enabling gradient updates for adaptation. G-Meta, proposed by Huang et al., combines ProtoNet and MAML, utilizing local subgraphs to transfer task information and enhance transfer efficiency. Meta-GNN [49], introduced by Zhou et al., integrates GNN to achieve cross-task generalization. These methods are

efficient but rely on task diversity. Metric learning-based methods learn class prototypes in the embedding space, supporting distance-based classification. The graph prototype network (GPN) [9] by Ding et al. extends PN to the graph domain, classifying through node importance estimation and prototype comparison. However, these methods are prone to overfitting when labels are scarce. Optimization strategy-based methods learn dynamic update rules to improve adaptability. Ravi et al. use LSTM as a meta-optimizer to adjust updates based on gradients and loss. These methods are robust but computationally complex. Although existing meta-learning methods effectively alleviate the problem of insufficient supervision information through task transfer [20], they still face challenges in extremely weakly supervised scenarios, where the lack of label information and insufficient task diversity remain limitations [6].

A.3 Task augmentation for solving few-shot problems

Despite the effectiveness of meta-learning frameworks in few-shot scenarios, the lack of task diversity remains a fundamental bottleneck. To address this, recent studies have proposed various task augmentation strategies. One category generates new tasks through data perturbation and mixing, such as GraphMix [23, 47], which interpolates node features, IA-FSNC, which enhances the support set and adds noise to features, and COSMIC, which mixes similar subgraphs to create "hard classes." Another category leverages self-supervision and pseudo-labeling to extract supervisory signals from unlabeled data, such as X-FNC, which expands supervision via pseudo-label propagation and the information bottleneck, and COLA [24], which combines contrastive learning with pseudo-label consistency to improve intra-class compactness and inter-class separability, though the quality of pseudo-labels can be unreliable under extremely weak supervision. A third category employs unsupervised or generative approaches, such as NAQ [17], which automatically generates diverse tasks to reduce distribution bias, and LLM4NG, which uses large language models to generate labeled nodes and edges, albeit with potential randomness and deviation from the original graph distribution. The fourth line of research focuses on graph foundation models [25], which learn transferable and general-purpose graph representations through large-scale pre-training. By leveraging massive unlabeled data and self-supervised objectives, these models effectively enhance generalization in few-shot scenarios, primarily providing support for downstream tasks at the representation learning level [52]. Overall, these methods are effective in enhancing task diversity but still face limitations regarding the semantic plausibility of perturbations, the reliability of generated signals, and fidelity to the original data distribution [16].

B Experimental Details

B.1 Analysis of Subgraph Construction

To evaluate the impact of the subgraph extraction module on model performance, we compare Personalized PageRank (PPR) and Random Walk strategies on the DBLP and Amazon-E datasets. As shown in Table 4, both methods achieve comparable classification accuracy across different few-shot settings, indicating that the proposed

Table 4: Comparison of PPR and Random Walk neighborhood construction on DBLP and Amazon-E (accuracy in %).

Dataset	DBLP				Amazon-E			
Setting	5-way 3-shot		10-way 3-shot		5-way 3-shot		10-way 3-shot	
# Labels per Class	10	20	10	20	10	20	10	20
PPR	81.4±1.2	82.1±2.8	68.7±3.2	69.4±2.9	77.4±2.5	76.3±1.8	66.3±3.4	69.9±2.6
Random Walk	80.8±2.4	81.2±2.6	67.8±2.6	69.1±3.0	74.5±2.8	73.1±3.2	65.7±3.3	65.4±2.6

Table 5: Comparison of Static and Dynamic Threshold Strategies on DBLP and Amazon-E datasets (accuracy in %).

Dataset	DBLP				Amazon-E			
Setting	5-way 3-shot		10-way 3-shot		5-way 3-shot		10-way 3-shot	
# Labels per Class	10	20	10	20	10	20	10	20
Static Threshold	81.4±1.2	82.1±2.8	68.7±3.2	69.4±2.9	77.4±2.5	76.3±1.8	66.3±3.4	69.9±2.6
Dynamic Threshold	80.2±2.4	81.6±3.0	68.9±2.6	69.6±2.4	79.2±3.1	79.5±2.8	65.3±2.2	71.2±3.2

framework is not sensitive to a specific subgraph extraction strategy.

This observation suggests that as long as the extracted subgraphs preserve key neighborhood information from the original graph, the model can maintain stable performance. In comparison, PPR consistently yields slightly better average accuracy, which may be attributed to its restart mechanism that favors neighbors with stronger structural relevance to the central node, thereby reducing noise. Consequently, we adopt PPR as the default subgraph construction strategy to provide more stable inputs for subsequent meta-learning and counterfactual generation.

B.2 Impact of Pseudo-Label Thresholding

To analyze the impact of different pseudo-label selection mechanisms on model training, we compare static and dynamic threshold

strategies on the DBLP and Amazon-E datasets. As shown in Table 5, the two approaches achieve comparable classification performance across various few-shot settings, with no significant performance difference observed. This result indicates that the performance gains of counterfactual meta-tasks mainly stem from effective selection of high-confidence samples rather than the specific form of the threshold, as long as counterfactual label flipping is applied only to nodes with high prediction confidence (i.e., $\max(\hat{y}_i) \geq \tau$), noise introduced by low-quality pseudo-labels can be effectively suppressed, ensuring the semantic reliability of the augmented tasks. Under this premise, a static threshold is sufficient to capture discriminative counterfactual information while avoiding the additional computational overhead of dynamic adjustment, validating the effectiveness and robustness of the proposed high-confidence selection mechanism.